

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE
AND EXPERT SYSTEMS

COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

*I **Kruthika Sre S / 192525360** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Kruthika Sre

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

1. PROBLEM ANALYSIS AND AGENT DESIGN

Problem Analysis

Universities receive a large number of students queries every day regarding course registration, class timetables, examination schedules, attendance records, and campus services. Students often have to wait for administrative staff to respond, especially during admission and examination periods. This increases staff workload and delays access to important information. Artificial Intelligence can solve this problem by using a conversational agent (chatbot). The chatbot uses Natural Language Processing (NLP) to understand user questions, identify the user's intent, and provide instant responses. It can operate continuously, reduce repetitive work for university staff, and improve the student support experience.

Agent Design

Component	Description
Agent Name	Student Support Assistant
Objective	To answer student queries related to registration, timetable, examinations, attendance, and campus services instantly and accurately.
Users	Students, prospective students, and university administrative staff.
Environment	University website, student portal, or mobile application.
Inputs	Student queries in natural language (text or voice).
Outputs	Relevant answers, guidance, or redirection to the appropriate department.
Performance Measures	Response accuracy, response time, user satisfaction, successful intent recognition, and handling of unknown queries.

2. DIALOGFLOW AGENT DEVELOPMENT

Create a Dialogflow ES agent named **Student Support Assistant**.

Intents

Intent	Purpose
Welcome	Greets the user
Course Registration	Answers registration-related questions
Timetable	Provides timetable information
Examination Schedule	Gives exam schedule details
Attendance	Responds to attendance-related queries
Campus Services	Provides information about campus facilities
Default Fallback	Handles unknown questions

Entities

Entity	Examples
@department	CSE, ECE, IT, Mechanical
@semester	Semester 1, Semester 2, Semester 3
@service	Library, Hostel, Transport, Cafeteria

Example Training Phrases

Course Registration

- How do I register for courses?
- Register my subjects
- Course registration process
- I want to enroll for this semester

Timetable

- Show my timetable
- Today's classes
- Class schedule

Examination

- When is my next exam?
- Exam timetable
- Upcoming examinations

Attendance

- What is my attendance?
- Attendance percentage
- Check attendance

Campus Services

- Where is the library?
- Hostel information

- Bus timings

Responses

Each intent should return a predefined response.

Example:

Timetable

Your class timetable is available on the student portal. Please select your department and semester to view the latest schedule.

Contexts

Example:

Course Registration → Registration Confirmation

If the user asks

How do I register?

then later asks

What documents are required?

Dialogflow uses context to understand that the second question still refers to course registration.

How Dialogflow Processes Queries

1. User enters a question.
2. Dialogflow performs Natural Language Processing.
3. The user's intent is identified.
4. Entities such as department or semester are extracted.
5. The matching intent is activated.
6. The chatbot generates the predefined response.
7. If no suitable intent is found, the Default Fallback Intent is triggered.

3. PROBLEM FORMULATION AND DECISION FLOW

Initial State

The chatbot is waiting for a student's query.

User Goal

Obtain correct information regarding university services.

Possible Actions

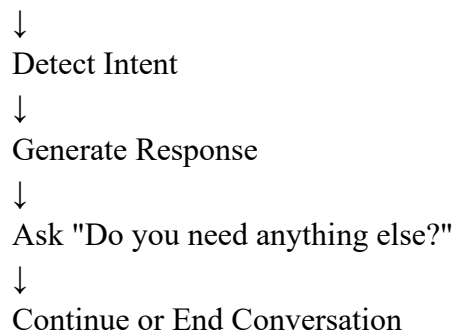
- Ask about registration
- Ask about timetable
- Ask about examinations
- Ask about attendance
- Ask about campus services
- Ask another question
- End the conversation

State Transitions

Initial State

↓

Receive User Query



Goal Completion Condition

The conversation ends after the chatbot successfully answers the student's query or directs the student to the appropriate department.

4. SEARCH STRATEGY APPLICATION

Conversational agents must determine the most appropriate response by selecting the correct intent based on the user's query. AI search techniques can support this process by efficiently exploring possible conversational paths.

Selected Search Strategy: Heuristic-Based Search

A heuristic-based search is appropriate because Dialogflow estimates which intent best matches the user's input using Natural Language Processing and confidence scores. Instead of examining every possible intent, the system prioritizes the most relevant intent based on similarity between the user's query and the training phrases.

For example, if the user enters:

"When are my semester exams?"

The chatbot compares the query with all available intents and selects the Examination Schedule intent because it has the highest matching confidence.

This approach improves response speed, reduces unnecessary comparisons, and provides more accurate results than searching every intent sequentially.

5. TESTING AND EVALUATION

User Query	Expected Intent	Result
Hi	Welcome	Correct
How do I register for courses?	Course Registration	Correct
Show my timetable	Timetable	Correct
What is my attendance percentage?	Attendance	Correct
Where is the library?	Campus Services	Correct
When are the exams?	Examination Schedule	Correct
Tell me a joke	Default Fallback	Correct

Evaluation

- Accuracy: The chatbot correctly identified the intended query for most test cases.

- Response Quality: Responses were clear, relevant, and easy to understand.
- Unknown Queries: Questions outside the chatbot's scope triggered the Default Fallback Intent with a polite response asking the user to rephrase or contact support.
- Limitations: The chatbot provides predefined responses and does not access live university databases.
- Future Improvements: Integrate the chatbot with student information systems for real-time attendance, timetable, examination schedules, and course registration details. Support multiple languages and voice-based interaction to improve accessibility.